

CFD Calculator

Standard earth gravity, g 9.80665 m/s²

Standard sea level conditions

Pressure, P 1.01325e5 Pa

Temperature, T 288.15 K

Boltzman constant, k_B 1.380649e-23 J/K

Gas constant, R 8.31446261815324 J/K mol

1. Hydraulic diameter, D_H

The hydraulic diameter is used to represent the equivalent diameter of fluid flow in tubes and channels that have non circular shape.

I. Annulus

$D_H = \text{Outer diameter} - \text{Inner diameter}$

II. Rectangular duct

$D_H = 2 * \text{width of the duct} * \text{height of the duct} / (\text{width of the duct} + \text{height of the duct})$

III. Others

$D_H = 4 * \text{cross sectional area of the flow} / \text{wetted perimeter of the cross section}$

Note: For fluid flow in a square duct, the length of the side is the hydraulic diameter.

2. Knudsen number, Kn

The Knudsen number helps decide whether to use statistical or continuum mechanics to model fluid flow in a specific situation.

$$Kn = k_B * T / (\sqrt{2} * \pi * d^2 * P * L)$$

Where, T – temperature, K

d – kinetic diameter, m

P – static pressure, Pa

L – characteristic length, m

Knudsen number	< 0.01	0.01 - 0.1	0.1 - 10	>10
Flow classification	Continuum	Slip flow	Transitional flow	Free molecular flow

3. Reynolds number, Re

The Reynolds number is used to characterise the fluids flow regime.

$$Re = \rho * u * L / \mu$$

Where, ρ – fluid density, kg/m³

u – freestream velocity, m/s

L – characteristic length, m

μ - dynamic viscosity, kg/m s

For flow in a pipe,

Reynolds number	<2000	2000 – 4000	>4000
Flow regime	Laminar	Transitional	Turbulent

4. Kinematic viscosity, ν

$$\nu = \mu / \rho$$

unit is m²/s

5. Hydrodynamic entrance length

For flow in a pipe, the length over which the flow evolves into fully developed velocity profile is called hydrodynamic entrance length.

I. For laminar flow

$$0.06 * Re * L$$

II. For transitional & turbulent flow

$$4.4 * Re^{(1/6)} * L$$

Where, Re – Reynolds number

L – characteristic length

6. Mach number, M

Mach number is the ratio of flow/object velocity to the speed of sound in the flow medium.

I. Speed of sound, m/s

a) For gases

$$\text{Speed of sound} = \sqrt{\gamma * R * \frac{T}{m}}$$

b) For liquids

$$\text{Speed of sound} = \sqrt{\frac{Ks}{\rho}}$$

II. M = Flow velocity / speed of sound

Where, γ – heat capacity ratio of gas

R – gas constant

T – Temperature, K

m – gas molar mass, kg/mol

Ks – liquid isentropic bulk modulus, Pa

ρ – liquid density, kg/m³

Mach number	<0.3	<1	1 – 5	>5
	Incompressible flow			
Speed range	Subsonic		Supersonic	Hypersonic

7. Rayleigh number, Ra

In a natural convection flow, Rayleigh number predicts whether the flow will be turbulent or not.

$$Ra = 2 * \rho^2 * L^3 * g * C_p * (T_w - T_\infty) / \mu * K * (T_w + T_\infty)$$

Where, ρ – gas density, kg/m³

L – characteristic length, m

g – standard earth gravity

C_p – gas specific heat capacity at constant pressure, J/Kg K

T_w – wall temperature (hot), K

T_∞ - bulk fluid temperature, K

μ - gas dynamic viscosity, kg/m s

K – gas thermal conductivity, W/m K

Note: This Rayleigh number calculation is based on ideal gas assumption.

8. Brinkman number, Br

Brinkman number helps determine if the viscous dissipation effects are significant in a given condition.

$$Br = \mu * u^2 / K * (T_w - T_\infty)$$

Where, μ - dynamic viscosity, kg/m s

u – freestream velocity, m/s

K – thermal conductivity, W/m K

T_w – wall temperature (hot), K

T_∞ - bulk fluid temperature, K

When $Br \geq 1$, viscous dissipation is significant.

9. Time step size, Δt

Estimate the time step size for a input courant number for a 3D simulation, based on mean cell size.

Time step size, Δt = desired courant (or) acoustic courant number * (fluid domain volume / number of fluid cells)^(1/3) / velocity

Where, fluid domain volume in m³

Velocity, m/s

I. For courant number

velocity = flow velocity

II. For acoustic courant number

velocity = flow velocity + speed of sound

Note:

- a) Assumed cells as cubes for the calculation.
- b) Speed of sound can be calculated from the mach number formula.

10. Drag Coefficient, Cd

Drag coefficient quantifies the drag of an object in the fluid.

$$Cd = 2 * Fd / (\rho * u^2 * A)$$

Where, Fd = force component in the flow direction, N

ρ – fluid density, kg/m³

u – velocity of object (relative to fluid), m/s

A – reference area, m²

11. Flow Coefficient, Cv

Calculate the flow coefficient of a device, which is a relative measure of its efficiency to allow liquid flow.

$$Cv = Q * \sqrt{\frac{\rho}{(P1 - P2) * 999.975}}$$

Where, Q – flow rate, US gallons/min

ρ – fluid density, kg/m³

P1 – upstream pressure, psi

P2 – downstream pressure, psi

12. First cell height

In a turbulent momentum boundary layer, Estimate the first cell height from a wall for the desired wall y^+

Reynolds number, $Re = \rho * u * L / \mu$

I. **First cell height** = $2^{1.5} * y^+ * L * (2\log Re - 0.65)^{1.15} / Re$

II. **Boundary layer total thickness** = $0.37 * L / Re^{(1/5)}$

III. **Number of cells**

a. If growth ratio is 1

= boundary layer total thickness / first cell height

b. If growth ratio is > 1

= $\log (1 - (\text{boundary layer total thickness} * (1 - \text{growth rate}) / \text{first cell height})) / \log (\text{growth rate})$

Where, y^+ - desired wall y^+

L – characteristic length, m

ρ – fluid density, kg/m^3

u – freestream velocity, m/s

μ - dynamic viscosity, kg/m s

Note: Ideal boundary layer growth rate should be ≤ 1.3

13. Prandtl number, Pr

Prandtl number gives the relative thickness of momentum and thermal boundary layers.

$$Pr = C_p * \mu / K$$

Where, C_p – specific heat capacity at constant pressure, J/Kg K

μ - dynamic viscosity, kg/m s

K – thermal conductivity, W/m K

Prandtl number	<1	1	>1
Thermal boundary layer thickness relative to momentum boundary layer	Thicker	same	thinner

References

1. Boundary-Layer theory by Dr.Herman Schlichting, 7th edition, 1979, ISBN 0-07-055334-3
2. Trends in Turbulence Treatments, Philippe R. Spalart, doi:10.2514/6.2000-2306
3. IAPWS, [International Association for the Properties of Water and Steam](#)
4. NIST, [Thermophysical Properties of Fluid Systems](#)

Fluid Properties

	Air	Water
Kinetic diameter d, m	3.64e-10	-
Density ρ , kg/m ³	1.225	999.102621467
Dynamic viscosity μ , kg/m s	1.7894e-5	1.13756755925e-3
Heat capacity ratio γ	1.4	-
Molar mass m, kg/mol	0.02801348	-
isentropic bulk modulus Ks, Pa	-	2.147020787517e9
specific heat capacity at constant pressure Cp, J/Kg K	1006.43	4188.46062263
Thermal conductivity K, W/m K	0.0242	0.588801733892

Unit conversion

Conversion factor to convert to m

	Mm	in	ft
Multiplication factor	1e-3	2.54e-2	0.3048

Conversion factor to convert to m²

	mm ²	in ²	ft ²
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Multiplication factor	1e-6	6.4516e-4	0.09290304
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Conversion factor to convert to K

°C	°F		From K to °F
$X + 273.15$	$(X + 459.67) * 5/9$		$X * 9/5 - 459.67$

Note: Temperature difference in °F, $\Delta T = (T_w - T_\infty) * 5/9$

Conversion factor to convert to Pa

	psi	bar	atm	Kpa
Multiplication factor	6894.75729316836	1e5	1.01325e5	1e3

Conversion factor to convert to kg/m³

	g/cm ³
Multiplication factor	1e3

Conversion factor to convert to kg/m s

	cP
Multiplication factor	1e-3

Conversion factor to convert to m²/s

	cSt
Multiplication factor	1e-6

Conversion factor to convert to m³

	mm ³	in ³	ft ³
Multiplication factor	1e-9	1.6387064e-5	0.028316846592

Conversion factor to convert to m³/s

	US gallons/min	lpm	cfm
Multiplication factor	6.30901964e-5	1.66666666666667e-5	4.719474432e-4

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